Fiscal Year:	FY 2019	**Task Last Updated:**	FY 03/29/2019
PI Name:	Kim, Jungho Ph.D.		
Project Title:	Gravity Effects on Flow Boiling Heat Transfer Using Local Heat Transfer Distribution Methods		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	kimjh@umd.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	301-405-5437
Organization Name:	University of Maryland		
PI Address 1:	Department of Mechanical Engineering		
PI Address 2:	2181 Glenn Martin Hall		
PI Web Page:			
City:	College Park	**State:**	MD
Zip Code:	20742-0001	**Congressional District:**	5
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	06/01/2015	**End Date:**	05/31/2020
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	McQuillen, John	**Contact Phone:**	216-433-2876
Contact Email:	john.b.mcquillen@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Colin, Catherine Ph.D. (Institut de Mécanique des Fluides de Toulouse, France)		
Grant/Contract No.:	NNX15AK16G		
Performance Goal No.:			
Performance Goal Text:			

EDITOR'S NOTE (5/17/2016): Continuation of "Measurement and Analysis of Gravity Effects on Flow Boiling Heat Transfer Using IR Thermometry" with same Principal Investigator, Dr. Jungho Kim, grant # NNX12AF31A. Added to Task Book when received information.
This effort will be conducted with the European Space Agency (ESA) in one of the facilities/test articles that they are developing.

Fluorescent paints that are sensitive to oxygen concentration (pressure sensitive paints, PSP) and temperature (temperature sensitive paints,TSP) have been used to measure shear stress, surface pressure, and wall temperature distributions in aerodynamic and fluid mechanics applications since the 1980s. It has seen very limited use for phase-change applications, however. If only temperature is to be measured, the paint is encapsulated within a binder or between impermeable barriers to eliminate sensitivity to oxygen. The molecules are typically stimulated using blue-UV

Task Description:	light, and emission at a longer wavelength occurs. The intensity of the emitted light decreases with temperature and oxygen concentration, and a filter is used to remove the excitation light before the emitted light is measured by a camera. The emission lifetime is typically on the order of microseconds, much faster than the boiling processes being measured, so the system frequency response is usually limited by the thermal response of the materials onto which the paint is bonded. The advantages of TSPs over infrared thermography and other techniques such as microheater arrays or microbolometers to measure temperature are numerous. Readily available CCD or CMOS cameras can be used to measure the intensity distribution instead of expensive IR cameras. Since TSPs respond very quickly, the transient response of TSP based measurements is limited in practice only by the speed of the camera, which is typically much faster than IR cameras. Also, since visible light is used for illumination and sensing, glass, clear plastics, PDMS (polydimethylsiloxane) and other commonly used materials can be used in place of expensive and difficult to machine IR transparent materials (e.g., silicon, sapphire, CaF2) as substrates. Calibration of TSPs can be performed without having to consider emission from the transparent substrate they are placed on, unlike for IR measurements. Finally, the wavelength of the light used (~0.5 mm for TSPs) is much smaller than the 3-5 µm or 8-12 µm light IR cameras are sensitive to, allowing much higher spatial resolution measurements. The diffraction limit for 0.6 µm light in air is ~0.3 µm. TSPs have quite high sensitivity (approximately -2%/°C), and TSPs are available that can be used over a wide temperature range. Although TSPs have been used for aerospace applications for many years, very little work has been performed using this technique for two-phase flows. We have developed methods whereby temperature and heat flux distributions can be measured during pool boiling and flow boiling so models of pool and flow boiling can be verified. A test apparatus has been developed and this technique will be used to measure gravity effects on flow boiling during a series of low-g aircraft flights in March, 2017.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The data is being used to develop and validate sub-models of the flow within each flow regime. Understanding of the heat transfer mechanisms will enable space-based heat exchangers to be designed with much more confidence. Although this work is primarily aimed at thermal management subsystems and development of lightweight evaporators, it also impacts other systems for novel power and transmission, advanced chemical propulsion, and energy storage.
Task Progress:	We obtained heat transfer data regarding flow boiling heat transfer in a 4 mm ID tube on the Zero-G aircraft (all of our previous work had been with 6 mm ID tubes) in November, 2018. Data were obtained at three subcoolings, 25-100 kg/m2-s flow rate, and various heat fluxes over four flights. Using a 4-way valve, data was obtained during upflow and downflow in 1.8 g (120 points), Martian gravity (12 points), Lunar gravity (6 points), and zero-g (112 points). We are in the process of reducing this data to obtain heat transfer coefficients along the tube, which will take a few months. Preliminary results indicated that as gravity decreases from 1.8 g to microgravity, the size of the bubbles increases, indicating less condensation around the bubble as the slip velocity (the difference in speed between the bubbles and the liquid) decreased. The mixing behind the bubble decreased as a result, so the heat transfer coefficient should also decrease. Very small bubbles occurred in the downward flow cases (negative g-values) due to the very high slip velocity, indicating high condensation and high heat transfer coefficient since the turbulence behind the bubbles mix the flow. If the bubbles fully condensed, however, the heat transfer is expected to decrease again.
Bibliography Type:	Description: (Last Updated: 03/13/2026)
Abstracts for Journals and Proceedings	Hammer CF, Lebon MT, Kim J. "Microgravity Flow Boiling Heat Transfer: Single-Phase Flow Simulations Using Local Liquid Velocity." 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. , Nov-2018
Articles in Peer-reviewed Journals	Lebon MT, Hammer CF, Kim J. "Gravity effects on subcooled flow boiling heat transfer." International Journal of Heat and Mass Transfer. 2019 Jan;128: 700-14. https://doi.org/10.1016/j.ijheatmasstransfer.2018.09.011 , Jan-2019